
DSC 40AR - Homework 7

due Tuesday, July 28th at 11:59PM

Homeworks are due to Gradescope by 11:59PM on the due date.

You may request an extension for this assignment by submitting [the form](#).

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should **always explain and justify** your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Homeworks should be written up and turned in by each student individually. You may talk to other students in the class about the problems and discuss solution strategies, but you should not share any written communication and you should not check answers with classmates. You can tell someone how to do a homework problem, but you cannot show them how to do it. **Only handwritten solutions will be accepted (use of tablets is permitted). Do not typeset your homework (using L^AT_EX or any other software).** Staff will not grade typed submissions.

For each problem you submit, you should **cite your sources** by including a list of names of other students with whom you discussed the problem. Instructors do not need to be cited.

Notes:

- For full credit, make sure to **assign pages to questions** when you upload your submission to Gradescope.
- Throughout this homework, it's fine to leave answers unsimplified, in terms of factorials, exponents, the permutation formula $P(n, k)$, and the binomial coefficient $\binom{n}{k}$.

Problem 1. Amped Up

Congratulations — you've won a month's supply of energy drinks! Each week, you receive **one prize box with 8 drinks** containing an assortment of drinks from companies like Monster, Red Bull, Celsius, etc. Each week's box is a surprise; you don't know which drinks it will contain until you open it.

For this problem, assume:

- There are 50 different possible energy drinks (equivalently, 50 different companies).
- Each of the 50 drinks is manufactured in equal quantities, so you are no more likely to get any one drink than any other.
- Each company produces only one type of energy drink, so (for instance) all drinks produced by Monster are the same, all drinks produced by Red Bull are the same, and so on.

Thus, saying that you have drinks from “different companies” is the same as saying that you have different *types* of drinks.

Note: Your final answers can include $P(n, k)$, $C(n, k)$, or $\binom{n}{k}$ without simplifying further.

- a) 🍷🍷🍷 Suppose that the contents of each week's prize box are selected uniformly and independently at random **without replacement** from among the 50 possible drinks (so each box contains 8 distinct drinks). If you take 2 weeks' worth of prize boxes (i.e., 16 total energy drinks), what is the probability that you end up with drinks from exactly 8 different companies?
- b) 🍷🍷🍷 Suppose that the contents of each week's prize box are selected uniformly and independently at random **without replacement** from among the 50 possible drinks. If you take 2 weeks' worth of

prize boxes (i.e., 16 total energy drinks), what is the probability that you end up with drinks from exactly 16 different companies?

- c) 🥑🥑🥑 Now suppose that the contents of each week's prize box are selected uniformly and independently at random **with replacement** from among the 50 possible drinks (so repeats are allowed, even within a single prize box). If you take 2 weeks' worth of prize boxes (i.e., 16 total energy *drinks*), what is the probability that you end up with drinks from exactly 16 different companies?
- d) 🥑🥑🥑🥑 Suppose that the contents of each week's prize box are selected uniformly and independently at random **with replacement** from among the 50 possible drinks. If you take 2 weeks' worth of prize boxes (i.e., 16 total energy drinks), what is the probability that you end up with drinks from exactly 15 different companies?

Problem 2. Combinatorial Proofs

A combinatorial proof is a method of proving that two mathematical expressions are equal by arguing that they both count the number of elements in the same set, i.e. they both count the number of ways to do the same thing. In **Lecture 7**, we gave a combinatorial proof of the fact that $\binom{9}{2} + \binom{9}{3} = \binom{10}{3}$; you may want to review that before proceeding.

Before we set you off to work on your own combinatorial proofs, we'll work through another example together. Here, we'll prove that:

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

A good first step is to verify yourself that the equation is true for some small values of n , both so that you understand how both sides of the equation are computed, and to start giving you an intuition for what both sides count. For example, when $n = 3$, the left-hand side is $\binom{3}{0} + \binom{3}{1} + \binom{3}{2} + \binom{3}{3} = 1 + 3 + 3 + 1 = 8$, which is indeed equal to $2^3 = 8$.

Here's how we'll proceed: we're going to argue that **both sides count the number of subsets of a set of n elements**. As an example, when $n = 3$, a set with n elements might look like $S = \{A, B, C\}$. A subset of S is any other set made up of elements from S . The 8 subsets of S are:

$$\{\}, \{A\}, \{B\}, \{C\}, \{A, B\}, \{A, C\}, \{B, C\}, \{A, B, C\}$$

Remember, our argument needs to work for any arbitrary positive integer n , not just when $n = 3$. So, let S be a set with n elements. A valid combinatorial proof is then:

- **Left-Hand Side:** A subset of S must either have 0 elements, 1 element, 2 elements, 3 elements, or so on, through n elements. There are $\binom{n}{0}$ subsets with 0 elements (this is just the empty set), $\binom{n}{1}$ subsets with 1 element, $\binom{n}{2}$ subsets with 2 elements, and so on. In general, there are $\binom{n}{k}$ subsets of S with k elements, since there are $\binom{n}{k}$ ways to select k elements from S without replacement such that order doesn't matter (which they don't in sets). So, the total number of subsets of S is $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = \sum_{k=0}^n \binom{n}{k}$, which is the left-hand side.
- **Right-Hand Side:** To construct a subset of S , for each of its n elements, there are 2 choices: include it in the subset, or do not include it in the subset. For element 1, there are 2 options; for element 2, there are 2 options; and so on. So, the total number of subsets of S is:

$$\underbrace{2 \cdot 2 \cdot \dots \cdot 2}_{n \text{ times}} = 2^n$$

which is the right-hand side.

This completes our combinatorial proof of the fact that $\sum_{k=0}^n \binom{n}{k} = 2^n$. Now, it's your turn!

- a) 🥑🥑🥑 Give a combinatorial proof of the fact that, that for positive integers m and n , and any integer k such that $0 \leq k \leq \min(m, n)$:

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \binom{m+n}{k}$$

Hint: Try and come up with a "story" in real-life terms, e.g. "Suppose we want to choose a committee of k students from ..."

- b) 🥑🥑 Evaluate the following expression for $n = 2$, $n = 3$, and $n = 4$, and show your work:

$$\sum_{i=0}^n \left(\binom{n}{i} \sum_{j=0}^{n-i} \binom{n-i}{j} \right)$$

Then, based on your evaluations for $n = 2$, $n = 3$, and $n = 4$, determine a simplified formula for the expression for a general value of n .

Hint: It should be a very simple formula.

- c) 🥑🥑🥑🥑 Give a combinatorial proof of your simplified answer from part (b). Your answer to (b) should have been a simple function $f(n)$, so you should be giving a combinatorial proof that:

$$\sum_{i=0}^n \left[\binom{n}{i} \sum_{j=0}^{n-i} \binom{n-i}{j} \right] = f(n)$$

Note that you won't receive any points for (c) if your $f(n)$ is incorrect. We're happy to confirm in office hours whether you have the right $f(n)$, but only if you show us your work in (b)!

Hint: Suppose I have n shirts and I'm deciding whether to keep, donate, or sell each one. How many ways can I decide what to do?

Problem 3. Ultrapermutations

A flight on Triton Airlines is scheduled for n passengers, all of whom have an assigned seat. Right before boarding, Triton Airlines' data systems malfunction and they lose track of the seat assignments. All n passengers board the flight but sit in a random seat. **What is the probability that nobody is sitting in their correct, originally assigned seat?** This is the question we will ultimately answer in this problem, but it'll take a few steps to get there.

To illustrate, let $n = 3$, and let's say our passengers are numbered 1, 2, and 3, where passenger i was originally supposed to sit in seat i . There are $3! = 6$ possible permutations of our 3 passengers, corresponding to the 6 ways in which they could've sat in the 3 seats on the plane:

$$123 \quad 132 \quad 213 \quad \boxed{231} \quad \boxed{312} \quad 321$$

For instance, 123 refers to the permutation where seat 1 is taken by passenger 1, seat 2 is taken by passenger 2, and seat 3 is taken by passenger 3, and 312 refers to the permutation where seat 1 is taken by passenger 3, seat 2 is taken by passenger 1, and seat 3 is taken by passenger 2.

Only 2 of the above permutations were ultrapermutations, where none of the 3 passengers were sitting in their originally assigned seat. (If you look at any permutation above that isn't boxed, at least one passenger is sitting in their correctly assigned seat.)

The probability that nobody is sitting in their correct, originally assigned seat is the probability of an ultrapermutation. For $n = 3$, there are 2 ultrapermutations out of 6 total permutations, so we'd say:

$$\mathbb{P}(\text{ultra}_3) = \frac{2}{6} = \frac{1}{3}$$

a) 🥑🥑 Let's work out one more concrete example. Let $n = 4$. Write out all $4! = 24$ permutations of four passengers, 1, 2, 3, and 4. How many of the 24 permutations are ultrapermutations? What is $\mathbb{P}(\text{ultra}_4)$?

b) 🥑 Moving forward, let n be any positive integer (that is, not necessarily 3 or 4).

Let A_i be the event that passenger i is sitting in their originally assigned seat. Determine the value of $\mathbb{P}(A_i)$. Your answer should involve n .

c) 🥑🥑 Suppose $1 \leq i < j < k \leq n$. Determine the values of $\mathbb{P}(A_i \cap A_j)$ and $\mathbb{P}(A_i \cap A_j \cap A_k)$. (As in the previous part, your answer should involve n .)

*Hint: $\mathbb{P}(A_i \cap A_j)$ is the probability that two different passengers **both are** sitting in their correct, originally assigned seat.*

d) 🥑🥑 Let C_w be defined as follows:

$$C_w = \binom{n}{w} \cdot \mathbb{P}(\text{a particular combination of } w \text{ passengers are all sitting in the correct seat})$$

Note that $\mathbb{P}(\text{a particular combination of 2 passengers are all sitting in the correct seat}) = \mathbb{P}(A_i \cap A_j)$ from part (c), and $\mathbb{P}(\text{a particular combination of 3 passengers are all sitting in the correct seat}) = \mathbb{P}(A_i \cap A_j \cap A_k)$ in part (c).

Show that:

$$C_w = \frac{1}{w!}$$

e) 🥑🥑 Explain why:

$$\mathbb{P}(A_1 \cup A_2 \cup \dots \cup A_n) = C_1 - C_2 + C_3 - C_4 + \dots C_n$$

Hint: Refer to Homework 6, Problem 2(c).

f) 🥑🥑 Show that:

$$\mathbb{P}(\text{ultra}_n) = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots \frac{1}{n!}$$

g) 🥑🥑 $\mathbb{P}(\text{ultra}_n)$ is equal to the sum of the first $n + 1$ terms of the [Taylor Series of \$e^x\$](#) , for a very specific value of $x = x_0$. What is x_0 ?

To show your work, you should write out the first few and last few terms of $\mathbb{P}(\text{ultra}_n)$ from part (f), the first few and last few terms of the Taylor Series of e^x in general, and the first few and last few terms of the Taylor Series of e^{x_0} , where x_0 is your chosen value of x .

h) 🥑 Determine the value of:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\text{ultra}_n)$$

This is the answer to the original question, “what is the probability that nobody is sitting in their correct, originally assigned seat?” for an arbitrarily large n .

Finding this limit shouldn’t require *any* work — it is a direct consequence of your answer to part (g). However, the purpose of this part is to have you take a step back and understand how all 8 parts of this problem tie together.

Problem 4. Baby Avi

When Avi was a baby, he was a picky eater. When he was offered a food, he'd eat it only sometimes, depending on the type of food. Baby Avi ate

- bananas 95% of the time,
- crackers 60% of the time,
- meat 45% of the time, and
- zucchini 30% of the time.

- a) 🥑🥑🥑 Baby Avi's grandpa gave him one of the above four foods and Baby Avi ate it all up. You have no idea which of the four foods it was, so assume it was equally likely to be any of them.

Given that Baby Avi ate the food, what's the probability that it was a banana? A cracker? Meat? Zucchini? Show your work.

- b) 🥑🥑🥑 As before, Baby Avi's grandpa gave him one of the above four foods and Baby Avi ate it all up. This time, suppose you know that Baby Avi's grandpa offers

- bananas 15% of the time,
- crackers 40% of the time,
- meat 10% of the time, and
- zucchini 35% of the time.

Given that Baby Avi ate the food, what's the probability that it was a banana? A cracker? Meat? Zucchini? Show your work.

- c) 🥑 Compare your answers to part (a) and part (b) above. Identify which of the four probabilities you computed increased and which decreased, and explain why this makes sense intuitively.

Problem 5. Independence and Conditional Independence

Consider the sample space $S = \{a, b, c, d, e, f\}$ with associated probabilities given in the table below.

outcome	a	b	c	d	e	f
probability	$\frac{4}{42}$	$\frac{4}{42}$	$\frac{2}{42}$	$\frac{10}{42}$	$\frac{16}{42}$	$\frac{6}{42}$

Let $X = \{a, b\}$ and $Y = \{b, e\}$. Remember to show your work for all calculations.

- a) 🥑 Are X and Y independent?

- b) 🥑🥑🥑🥑 In this problem, you will determine whether X and Y are conditionally independent given a third event, Z .

1. Suppose $Z_1 = \{a, b, d, e, f\}$. Are X and Y conditionally independent given Z_1 ?
2. Suppose $Z_2 = \{a, b, e\}$. Are X and Y conditionally independent given Z_2 ?
3. Suppose $Z_3 = \{c, d, e, f\}$. Are X and Y conditionally independent given Z_3 ?